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Long-Term Persistence of Hormonal Adaptations to Weight Loss

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ABSTRACT & BACKGROUND

Background: High rates of weight regain after successful diet-induced weight loss represen
t a persistent metabolic challenge.

This investigation evaluated whether changes in circulating levels of appetite-regulating hormones persist 1 year following weight reduction.

Methods: Fifty overweight or obese patients underwent a 10-week weight loss program with a very-low-energy diet.

Levels of nine circulating appetite hormones and subjective appetite scores were assessed at baseline, 10 weeks, and 62 weeks.

RESULTS: THE BIOLOGY OF THE HUNGER CRASH

Key Finding: Circulating levels of the hunger-stimulating hormone ghrelin remained signifi cantly elevated 1 year after initial weight loss.

Conversely, satiety hormones including leptin, peptide YY, and cholecystokinin remained si gnificantly suppressed compared to baseline.

Mean body weight loss was 13.5 kg at 10 weeks; participants had regained 5.5 kg by week 62 , yet hunger scores remained elevated.

Subjective hunger and desire to eat were significantly higher at week 62 than before the w eight-loss intervention began.

These persistent endocrine changes demonstrate that the post-diet hunger drive is a powerf ul biological mechanism to restore fat stores.

CLINICAL IMPLICATIONS FOR METABOLIC CARE

These findings prove that appetite increases and mid-afternoon hunger crashes are driven b y biology rather than personal weakness.

Because endocrine adaptations persist for at least one year, long-term multi-disciplinary management is essential to prevent regain.

Study Limitations: Small cohort of 50 patients; dietary compliance during the unsupervised maintenance period varied.

Conclusion: Counteracting persistent hormonal hunger adaptations requires continuous clini cian support and comprehensive lifestyle restructuring.